

Lab 3 ticket

Transforming Space — Friday, October 9

Name: _____ Language (circle): Octave / Python

New syntax (fill as you go)

I needed to...	Command / pattern (your language)
Enter a 2×2 matrix	
Compute Ax or $A * X$	
Plot 2D points	
diag / identity	

Store 2D points as **columns** of a matrix. Quiz 5 today is basis (not this lab). Quiz 6 will cover Week 7.

E1. $A = \text{diag}(2, 1/2)$, $x = (1, 1)$. Predict Ax , then check.

Expected: $(2, 0.5)$. **Yours:** prediction _____ code _____

Geometry in one phrase: _____

E2. Reflection through x_2 -axis: $e_1 \mapsto (-1, 0)$, $e_2 \mapsto (0, 1)$. Write A . Apply to columns of $X = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$.

$A =$ _____ $AX =$ _____

Expected: $A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$; $AX = \begin{bmatrix} -1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}$.

E3. Shear $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ on unit-square columns $X = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$. Compute $Y = AX$.

Expected: $Y = \begin{bmatrix} 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$. **Yours:** _____

What happens to $e_2 = (0, 1)$? _____

E4. 90° CCW: $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ on the same X . Compute Y ; note if it stays a square.

Expected: $Y = \begin{bmatrix} 0 & 0 & -1 & -1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$; yes. **Yours:** _____

E5. Adapt: use 180° rotation ($A = -I$) on X from E3. Where does $(1, 0)$ go?

Expected: $(-1, 0)$. **Yours:** _____ Command changed: _____

Stuck on (optional): E _____

Instructor initials: _____